EDITORIAL

Viewing the Value of Coronary Artery Calcium Testing From Different Perspectives

Tamar S. Polonsky, MD, MSCI; Philip Greenland, MD

Risk assessment is the foundation of atherosclerotic cardiovascular disease (ASCVD) prevention in the clinic. Treatment decisions for statins, antihypertensive medication, and aspirin are all based on 10-year estimated risk using risk factor equations.[1-3] Yet it has long been appreciated that risk scores are imperfect. A large proportion of ASCVD events occur among adults who are below the cutoffs used to initiate medical therapy, and many patients who do cross the threshold used to justify lifelong preventive medications are not destined to experience an ASCVD event.

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This week in *JAMA*, the US Preventive Services Task Force (USPSTF) provides recommendations[4] and an evidence report[5] on the potential use of 3 markers of subclinical CVD to enhance cardiovascular risk assessment in asymptomatic adults: the ankle-brachial index (ABI), high-sensitivity C-reactive protein (hs-CRP) level, and coronary artery calcification (CAC) score. The USPSTF stated that it chose the 3 markers because they have the most promising evidence base, are reliably measured, and are independently associated with cardiovascular events and also because the prevalence and distribution of abnormal and normal values have been described. However, the USPSTF stated that there was insufficient evidence to recommend for or against routine clinical use of any of the markers, citing what it determined were the following gaps in evidence: (1) the clinical meaning of any improvement in risk assessment with the markers was unknown, (2) treatment decisions guided by the markers have not been shown to reduce cardiovascular events or mortality, and (3) evidence on the addition of the markers to the Pooled Cohort Equations (PCE) is sparse.[4,5]

By discussing the ABI, hs-CRP, and CAC together, the USPSTF statement[4] suggests that the 3 markers are comparable in their abilities to improve risk assessment. However, as outlined in the accompanying evidence report,[5] studies directly comparing the ABI, hs-CRP, and CAC demonstrate that the use of CAC is far more likely than ABI or hs-CRP to provide meaningful alterations in patients' estimated risk.[5-7] The USPSTF acknowledged in its "Response to Public Comment" that many of the comments provided evidence that risk assessment with CAC was strong enough to warrant a separate, more positive recommendation. However, the USPSTF maintained that the clinical meaning of an improvement in risk assessment was unknown.

The USPSTF based its conclusions regarding the first gap in evidence on several statistical measures, such as the area under the receiver operator characteristic curve. The area under the receiver operator characteristic curve is a measure of discrimination: whether test results will differ between broad populations of adults who do and do not experience events. For example, the USPSTF cited a study using data from the Multi-Ethnic Study of Atherosclerosis (MESA) by Yeboah et al,[8] in which the authors used the PCE to estimate 10-year ASCVD risk and stratified the entire cohort into those with less than 7.5% or at least 7.5% risk.[8] Adding CAC to the model only improved the area under the receiver operator characteristic curve by 0.02 to 0.04. The USPSTF used these results as evidence that CAC only provides modest improvements in risk prediction and thus should not be used routinely. We agree with the recommendations that CAC testing should not be considered as a routine screening test for all asymptomatic individuals regardless of their traditional risk factors. In particular, CAC results are less informative when they are applied to individuals deemed to be at high or low risk by risk scores such as the PCE or Framingham Risk Score. Many low-risk individuals remain low risk, and many high-risk individuals remain high risk, even with CAC assessment. Coronary artery calcium testing is most helpful when used selectively for patients in whom treatment decisions are unclear.

The USPSTF acknowledged that measures of risk reclassification more closely reflect how clinicians might incorporate an additional risk marker into treatment decisions. In the evidence report, the authors focused on the same study by Yeboah et al[8] that evaluated CAC testing in the entire MESA cohort. The study showed that a large proportion of the adults who experienced events were appropriately reclassified to higher risk based on their CAC scores; however, among the adults who did not experience an event, more were inappropriately classified to higher risk than to lower risk. Again, the USPSTF's conclusion was that the improvement in risk prediction with CAC testing was too modest to recommend its use.

However, another study from MESA by Nasir et al[9] showed that CAC testing could provide meaningful prognostic information for a more focused population of adults whose levels were just greater than the 7.5% risk threshold. The authors found that 41% of adults who would have been recommended statins based on a 10-year ASCVD risk of at least 7.5% had a CAC score of 0. More specifically, in the 7.5% to 9.5% group (about one-quarter of statin-recommended MESA participants), the prevalence of a CAC score of 0 was even higher at 55%, and the event rate was quite low at 2.7 per 1000 person-years. The results suggest that the use of CAC testing could help identify a large proportion of adults who are otherwise statin-eligible in whom

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long-term drug therapy might safely be avoided. A 2018 MESA study confirmed the low 10-year event rate among men and women with CAC scores of 0.[10] Further, in the study by Nasir et al,[9] 17% of the adults with an estimated 10-year risk of 7.5% to 9.9% and 24% of the adults with a 10-year estimated risk of 10% to 14.9% had CAC scores greater than 100. The 10-year event rates for those with CAC scores greater than 100 were greater than 7.5%, regardless of sex or race/ethnicity, suggesting they are more likely to benefit from medical therapy.[10] The authors of the evidence report stated that they did not include the study by Nasir et al[9] because of several differences in design between that study and the one by Yeboah et al,[8] precluding direct comparisons. However, the 2 studies are complementary. Taken together, the results highlight that CAC testing is most valuable for the many adults at the borders of current treatment thresholds and to clarify decisions about whether to start or to intensify therapy.

We concur with the second gap in evidence cited by the USPSTF regarding lack of clinical trial data demonstrating a reduction in incident CVD events or mortality with CAC testing. However, the strongest design of an outcome-driven randomized clinical trial would dictate that adults with a CAC score of 0 forgo statin therapy and only those with elevated CAC scores would be treated.[11] Such a study requires a sample size of more than 20 000 participants with long follow-up and would not likely be funded.[11] To expect outcome data is too narrow an approach given the strength of the observational data for CAC testing in selected risk groups.

As an alternative for risk assessment, the USPSTF recommended using the PCE as the best approach to clinical decision making despite the lack of outcome data validating its use or any other risk score. Since the PCE was published in 2013, there has been substantial discussion about whether or not the PCE overestimates risk.[12,13] The third concern raised by the USPSTF is that most studies added CAC to risk equations using the same variables as the PCE but were recalibrated to the cohorts being studied. The USPSTF suggested that a more meaningful approach would have been to add CAC to the published PCE, presumably because this would reflect how clinicians might apply CAC results in clinical care. However, if CAC is studied using a risk score that is not well calibrated to the population, then any improvement in risk prediction with CAC will be amplified. The approach favored by the USPSTF would likely point to an even higher yield with CAC testing than what has been seen in most published studies.

Discussions of cost-effectiveness and shared decision making are not included in the guidelines because the USPSTF does not consider costs of providing a service.[4] However, cost-effectiveness analyses can be valuable clinical tools that help guide discussions with patients about how to balance the risks of testing, treatment, and clinical events with more patient-centered factors such as concerns about adverse effects or the desire to avoid a lifelong medication. Hong et al[14] found that the economic value of a CAC-guided treatment strategy was similar to a strategy in which all patients who were eligible for statin therapy were treated. However, the results were highly dependent on assumptions associated with the cost of CAC testing and medication and the disutility associated with taking a statin.

Shared decision making between the clinician and patient is particularly important when implementing strategies for primary prevention, a scenario in which there is inherent uncertainty about the yield of treatment.[1,15] Guidelines emphasize that estimated ASCVD risk using the PCE is not meant to serve as an absolute determination of who does and does not qualify for statin therapy; rather, it is meant to frame a conversation between the patient and clinician.[1] Many patients may require more than risk estimates alone to reassure them that medical therapy can either be safely withheld or should be used long term to alter the trajectory of ASCVD. The USPSTF acknowledged that the ultimate outcome of potential harms associated with CAC testing, such as radiation exposure and incidental findings, is unknown. Downstream testing can be minimized if patients and clinicians understand that CAC testing should not be used to justify additional procedures such as stress testing or a coronary angiogram. In conclusion, in our view, there is more than sufficient evidence to support selective use of CAC testing as a tool to inform risk assessment and support the less tangible aspects of clinical decision making.

ARTICLE INFORMATION

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Decoding the Right Ventricle in 3 Dimensions

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In 1632, Galileo Galilei wrote in his *Dialogo Sopra I Due Sistemi Massimi del Mondo* (*Dialogue Concerning the Two Chief World Systems*), "All truths are easy to understand once they are discovered; the point is to discover them." This quote by one of the world's most famous scientists states that some truths need to be discovered before they can be widely accepted. In this issue of *JAMA Cardiology*, Nochioka et al[1] sought to study the prognostic significance of a variety of right ventricular (RV) parameters, including 3-dimensional (3-D) echocardiographic assessment of the right ventricle in an elderly, population-based cohort. The authors found an association between RV function assessed by 3-D imaging and heart failure (HF) outcomes in this population. The results of this study underscore the importance of including 3-D measurements in the comprehensive assessment of the right ventricle in the echocardiography laboratory.

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Although previously thought of as a passive conduit or a so-called bystander to the left heart, the dysfunctional right ventricle has recently been shown to have considerable impact on the morbidity and mortality of patients with cardiovascular disease. Currently, functional assessment of the right ventricle on echocardiography is largely based on qualitative measures, although quantification of RV size and function using indices described in current guidelines[2] have been shown to correlate well with the gold standard of cardiovascular magnetic resonance.[3]

In this study by Nochioka et al,[1] 2-dimensional (2-D) and 3-D imaging of the right ventricle were performed at the fifth visit in participants enrolled in the Atherosclerosis Risks in the Community study. After categorizing participants per the American College of Cardiology/American Heart Association stages of HF, the authors found that RV systolic function as well as RV-pulmonary artery coupling progressively declined across all stages.[1] Interestingly, RV function was impaired in approximately one-fourth of asymptomatic people with stage B HF. As expected, RV ejection fraction (EF) correlated inversely with afterload and directly with left ventricular (LV) function. Among patients free of HF, both worse RVEF and RV-pulmonary artery coupling were linearly associated with increased risk of HF or death independent of LVEF, N-terminal pro brain natriuretic peptide levels, and measures of LV filling pressure. Interestingly, these findings did not extend to 2-D RV free-wall longitudinal strain, RV fractional area change, and tricuspid annular systolic velocity. Notably, tricuspid annular plane systolic excursion, an important 2-D measure of RV function with well-documented prognostic value, was not reported in this study. Additionally, RV basal and mid-ventricular dimensions, which are commonly used in clinical practice, were not included.

The right ventricle is a complex, crescent-shaped chamber composed of 3 segments: the inflow tract, the body, and the outflow tract. To evaluate the entire right ventricle on 2-D imaging, multiple views are required. Even so, when the right ventricle remodels in volume or pressure overload states, its morphology alters considerably, so that it is next to impossible to completely assess its shape from 2-D imaging alone.[4] This explains why 2-D echocardiographic assessment of RV size correlates less robustly with cardiovascular magnetic resonance–derived RV volumes especially in larger right ventricles.[5] On the other hand, 3-D RV volumes have been shown to correlate well with cardiovascular magnetic resonance in multiple studies.[5] Because it has been shown that cardiovascular magnetic resonance RV volumes have prognostic significance in multiple disease states, it would have been of interest to learn about the changes in 3-D RV volumes with progressive stages of HF.

It has been our experience that 3-D RV acquisition and analysis has a steep learning curve. In this study,[1] the authors report a very impressive feasibility for 3-D RV acquisition and analysis, with optimal image quality obtained in 1004 of 1225 participants (82.0%). It would be interesting for the echocardiographic community to understand how this was achieved. Several questions come to mind. Was training in data acquisition conducted beforehand? How many sonographers participated in 3-D imaging acquisition? Was feasibility similar across sites? This result is highly commendable and the methodology should be shared to improve

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